

## QUOTATION



QUOTE Update: 2015-05-27

PROJECT NO.: HC159070

QUOTATION NO.: QC1560705 Rev.1

CUSTOMER: Hyundai WIA Zhangjiagang

### **AUTOMATIC INDUCTION HARDENING & TEMPERING MACHINE FOR OUTER RACE**

Machine structure: linear type machine with in-stocker (chain type conveyors + turn table lifter), washing station, process machine with bell and stem hardening stations, eddy-current test station, marking station (dot marking + laser marking), tunnel tempering station, out-conveyor and a pick-up & place loader

- Machine type: HARDLINE TS 350.3
- Power source for hardening: SINAC 150/240 SM DUAL
- Power source for tempering: SINAC 50 PM
- Control system: SIEMENS 840D SL
- NC axis-Z1 at the bell hardening station
- NC axis-Z2 at the stem hardening station
- Parts to be heat treated: outer race

## 1.1. SCOPE OF SUPPLY & PRICE

### 1.1.1 SCOPE OF SUPPLY

| ITEM | QTY.      | NAME                                                                                            | TYPE & ORIGINAL MANU.                 |
|------|-----------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| 1    | 1 set     | IGBT power source for hardening                                                                 | SINAC 150/240 SM DUAL<br>(EFD Norway) |
| 2    | 1 set     | IGBT power source for tunnel tempering                                                          | SINAC 50 PM<br>(EFD Induction)        |
| 3    | 1 set     | Process machine with bell hardening station and stem hardening station                          | HARDLINE TS 350.3<br>(EFD China)      |
| 4    | 1 set     | Tunnel tempering machine excluding tempering coils                                              | (EFD China)                           |
| 5    | 1 set     | CNC control system                                                                              | Siemens 840D SL<br>(Siemens Germany)  |
| 6    | 1 set     | Cooling unit for quench medium                                                                  | RA-600.2000.300x2<br>(EFD China)      |
| 7    | 1 set     | Cooling unit for cooling water                                                                  | RC-400MJ-6-200<br>(EFD China)         |
| 8    | 1 set     | Chiller, water cooled type                                                                      | 50kW<br>(EFD China)                   |
| 9    | 1 set     | Quality monitoring system                                                                       | QMS<br>(EFD Germany)                  |
| 10   | 14 pieces | Bell hardening coil according to WIA specification                                              | EFD<br>(EFD China)                    |
| 11   | 10 pieces | Stem hardening coil according to WIA specification                                              | EFD<br>(EFD China)                    |
| 12   | 2 sets    | Tunnel tempering coils according to WIA specification                                           | EFD<br>(EFD China)                    |
| 13   | 7 sets    | Jigs according to WIA specification                                                             | EFD<br>(EFD China)                    |
| 14   | 1 set     | washing machine                                                                                 | Korean brand                          |
| 15   | 1 set     | In-stocker: chain type conveyor with turn table lifter for max. 200 parts (workpiece dia. 89mm) | EFD China                             |
| 16   | 1 set     | Out conveyor with a pick-up & place loader for max. 20 parts (workpiece dia. 89mm)              | EFD China                             |

|    |       |                                                                                                                                           |                    |
|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 17 | 1 set | Dot marking equipment                                                                                                                     | JEIL               |
| 18 | 1 set | Laser marking equipment                                                                                                                   | PANASONIC          |
| 19 | 1 set | EDDYVISOR DS Model, 2 channels, incl. 14x eddy coils for bell and 4x eddy coils for stem                                                  | IBG                |
| 20 |       | Pre-acceptance at EFD Shanghai                                                                                                            | EFD<br>(EFD China) |
| 21 |       | Packing, truck transportation and insurance to WIA Zhangjiagang                                                                           | EFD<br>(EFD China) |
| 22 |       | Installation, commissioning and final acceptance at WIA Zhangjiagang                                                                      | EFD<br>(EFD China) |
| 23 | 1 set | Spare parts<br>(WIA will select spare parts valued Korean WON 40,000,000 from a recommended list submitted by EFD after designs approval) | EFD<br>(EFD China) |
| 24 |       | Warranty period: 12 months after final acceptance (2 years for SIEMENS PLC only)                                                          | EFD<br>(EFD China) |

### 1.1.2 PRICING

TOTAL PRICE FOR ALL THE ITEMS ABOVE:

**7,550,000.--RMB (Excl. 17% VAT)**

### 1.2. CYCLE TIME

- Cycle time: 25 seconds per piece incl. tempering

### 1.3. UTILITIES IN THE CUSTOMER'S WORKSHOP

- **MAINS VOLTAGE AND CAPACITY:**
  - ✧ 3 phases, 380V -10%/+10%, 50Hz; 480 KVA
- **PLANT COOLING WATER:**
  - ✧ Flow rate: 800litre/min.; water pressure: 2.5 - 4 bar; temperature: ≤28℃
- **COMPRESSED AIR:**
  - ✧ Pressure: 4-6 bar; consumption: 200l/min.

## 2. System Survey

### **AUTOMATIC INDUCTION HARDENING & TEMPERING MACHINE, TYPE HARDLINE TS 350.3**

| Item | Qty. | Description |
|------|------|-------------|
|------|------|-------------|

---

#### **2.1 Power Source for hardening**

|       |   |                                                                                      |
|-------|---|--------------------------------------------------------------------------------------|
| 2.1.1 | 1 | <b>IGBT Transistorized Frequency Converter</b><br><b>Type: SINAC 150/240 SM DUAL</b> |
|-------|---|--------------------------------------------------------------------------------------|

|                      |     |    |
|----------------------|-----|----|
| Nominal output power | 150 | kW |
|----------------------|-----|----|

Max. output power

|                     |     |    |
|---------------------|-----|----|
| (Intermittent duty) | 240 | kW |
|---------------------|-----|----|

|                        |        |     |
|------------------------|--------|-----|
| Duty factor/Cycle time | 0.5/10 | min |
|------------------------|--------|-----|

|                        |       |     |
|------------------------|-------|-----|
| On station I for bell: | 10-25 | kHz |
|------------------------|-------|-----|

|                         |      |     |
|-------------------------|------|-----|
| On station II for stem: | 9-13 | kHz |
|-------------------------|------|-----|

The frequency range changing will be selected manually on station II

#### **Features and Modules:**

- compact cabinet for indoor operation (IP 54 protection)
- mains input terminal
- diode rectifier
- converter with IGBT-transistors
- output capacitor module
- electronic control device
- indirect water cooling system
- external or internal power pre-selection / regulation
- LED signal fault display (alarm card)

|       |   |                                                                                                                        |
|-------|---|------------------------------------------------------------------------------------------------------------------------|
| 2.1.2 | 2 | High current transformer type HHT incl. flexible connecting cable between converter exit and transformer, water cooled |
|-------|---|------------------------------------------------------------------------------------------------------------------------|

|       |   |                                                                   |
|-------|---|-------------------------------------------------------------------|
| 2.1.3 | 1 | Automatic change-over switch to switch the power from two outputs |
|-------|---|-------------------------------------------------------------------|

|       |   |                                                                   |
|-------|---|-------------------------------------------------------------------|
| 2.1.4 | 2 | Set of inductor connecting brackets, quick clamping type (manual) |
|-------|---|-------------------------------------------------------------------|

| Item       | Qty. | Description                                                                                                                                                                                                                                                                                                                                                                       |
|------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2.2</b> |      | <b>Power Source for tunnel tempering</b>                                                                                                                                                                                                                                                                                                                                          |
| 2.2.1      | 1    | <b>Frequency Converter</b><br><b>Type: SINAC 50 PM</b>                                                                                                                                                                                                                                                                                                                            |
|            |      | Output power 50 kW<br>Frequency range: 1 - 3 kHz                                                                                                                                                                                                                                                                                                                                  |
|            |      | <b><u>Features and Modules:</u></b> <ul style="list-style-type: none"> <li>• compact cabinet for indoor operation (IP 54 protection)</li> <li>• diode rectifier</li> <li>• converter</li> <li>• output capacitor module</li> <li>• electronic control device</li> <li>• indirect water cooling system</li> <li>• external or internal power pre-selection / regulation</li> </ul> |

| Item       | Qty. | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2.3</b> |      | <b>Process Machine</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.3.1      | 1    | <b>Automatic hardening &amp; tempering machine</b><br><b>Type: HARDLINE TS 350.3</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|            |      | <b><u>Features and Modules</u></b> <ul style="list-style-type: none"> <li>• machine frame in rigid and stainless steel design with integrated quench ant collecting basin (stainless steel)</li> <li>• roll away covers to protect the mechanical assemblies</li> <li>• working area lamp</li> <li>• vented hood with air purifier (brand: NEDEMANN)</li> <li>• automatic central lube system with pressure supervision, CNC monitored</li> <li>• inductor cooling water flow control (min.) with manual shut on/off-valve</li> <li>• quenchant lines with shut off valve, strainer, manometer, electro-pneum. activated valve (on/off) and manual setting valve</li> <li>• transformer/inductor carriage (aluminium cast)</li> <li>• quench barrel holding device</li> <li>• bell hardening station: One vertical NC-Z1-axis for part</li> </ul> |

|        |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |   | <p>movement from top to bottom</p> <ul style="list-style-type: none"> <li>• bell hardening station: coil movements forward/back/left/right are adjusted manually, with position sensors and the movement values are displayed on the screen</li> <li>• stem hardening station: vertical NC-Z2-axis for part movement up/down</li> <li>• stem hardening station: coil movements forward/back/left/right are adjusted manually, with position sensors and the movement values are displayed on the screen</li> <li>• Automatic part-transfer will be controlled by NC servo</li> <li>• 1x motorised aluminium cast tailstock, which can be set at different height for changing admit between centres at the stem hardening station</li> <li>• 1x pyrometer to measure part surface temperature after tempering but before cooling</li> <li>• pneumatic service unit</li> <li>• side assembly of machine control cabinet (water cooling type air conditioner, RITTAL) with implemented panel for operating and indicating instruments at front end</li> <li>• CNC/PLC-system Type SIEMENS 840D SL 10,4" colour screen</li> <li>• pilot lamps (Green, Yellow, Red)</li> </ul> |
| 2.3.2  | 1 | Tunnel tempering machine with NG chute (excl. tempering coils, but one set installed on the tunnel tempering machine and one spare set are shown below)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2.3.3  | 1 | In-stocker: chain type conveyor with turn table lifter for max. 200 parts (workpiece dia. 89mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.3.4  | 1 | Out conveyor with a pick-up & place loader for max. 20 parts (workpiece dia. 89mm)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2.3.5  | 1 | Automatic part-transfer from the inlet to the outlet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2.3.6  | 1 | Washing machine                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.3.7  | 1 | Dot marking equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.3.8  | 1 | laser marking equipment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2.3.9  | 1 | Eddy current detector (IGB Eddyvisor DS Model, 2 channels) with NG reject bin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.3.10 | 1 | An idle station between bell hardening station and stem hardening station                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Others

- |        |   |                                                                                                                                                                                              |
|--------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.3.11 | 1 | Booster pump for inductor cooling water for higher cooling efficiency and longer inductor life time<br>(10 - 50 l/min., 7,5 bar)                                                             |
| 2.3.12 |   | Mains distribution cabinet incl. main switch, with fused exits for power source, process machine and cooling unit, control for cooling unit and protection switch for circulating pump motor |

## Quality monitoring system comprising:

- |        |   |                                                                                                                                                                                                                                                                                  |
|--------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.3.13 | 2 | Electronical energy counter with pre-selectable min./max. limit values to control the actual energy transferred into the work piece per heating cycle, decentralized fault messages in clear text on machine control panel                                                       |
| 2.3.14 | 4 | Inductive quench medium (2 for bell station and 2 for stem station) flow supervision with digital display of actual value, various min./max. values pre-selectable via CNC, pre-setting of actual value manually by valve, fault message in clear text on machine control panel. |
| 2.3.15 | 1 | Quenchant temperature supervision with digital display and electrical contacts for min./max. settings, fault message decentralized in clear text on machine control panel                                                                                                        |
| 2.3.16 |   | Power sources, process machines, cooling unit and electrical mains cabinet completely assembled incl. Pipe work and wiring, functionally tested                                                                                                                                  |

|      |      |             |
|------|------|-------------|
| Item | Qty. | Description |
|------|------|-------------|

---

## 2.4 Cooling systems

- |       |   |                                                                                                  |
|-------|---|--------------------------------------------------------------------------------------------------|
| 2.4.1 | 1 | <b>Cooling Unit for Quench Medium<br/>(heat exchanger principle)<br/>Type: RA-600.2000.300x2</b> |
|-------|---|--------------------------------------------------------------------------------------------------|

### Comprising:

Water tank in stainless steel, circulating pump with motor (brand: NANFANG), manometer, thermometer, plate type heat exchanger, electric heater, magnet separator, oil skimmer, paper filter, etc.

### **Technical Data**

|                  |             |
|------------------|-------------|
| Cooling capacity | 600 MJ/h    |
| Tank capacity    | 2000 l      |
| Pump flow rate   | 300x2 l/min |
| Pump pressure    | 7 bar       |

- 2.4.2      **1      Cooling Unit for cooling water  
(heat exchanger principle)  
Type:    RC-400MJ-6-200**

#### **comprising:**

Single filter with metal strainer, circulating pump with motor (brand: NANFANG), manometer, thermometer, plate type heat exchanger etc.

### **Technical Data**

|                  |                                 |
|------------------|---------------------------------|
| Cooling capacity | 400 MJ/h                        |
| Tank capacity    | utilize the tank in the chiller |
| Pump flow rate   | 200 l/min                       |
| Pump pressure    | 6 bar                           |

- 2.4.3      **1      Chiller, water cooled, cooling capacity 50kW, tank size  
1000L**



| Item       | Qty. | Description          |
|------------|------|----------------------|
| <b>2.5</b> |      | <b>Process Tools</b> |

|       |        |                                                      |
|-------|--------|------------------------------------------------------|
| 2.5.1 | 7 sets | Jigs related to outer races in the WIA specification |
|-------|--------|------------------------------------------------------|

| Item       | Qty. | Description  |
|------------|------|--------------|
| <b>2.6</b> |      | <b>Coils</b> |

|       |        |                                                                             |
|-------|--------|-----------------------------------------------------------------------------|
| 2.6.1 | 14 pcs | Coil in the first version for hardening bell according to WIA specification |
|-------|--------|-----------------------------------------------------------------------------|

|       |        |                                                                                                    |
|-------|--------|----------------------------------------------------------------------------------------------------|
| 2.6.2 | 10 pcs | Coil with attached quenches in the first version for hardening stem according to WIA specification |
|-------|--------|----------------------------------------------------------------------------------------------------|

|       |        |                                                                                |
|-------|--------|--------------------------------------------------------------------------------|
| 2.6.3 | 2 sets | Coils in the first version for tunnel tempering according to WIA specification |
|-------|--------|--------------------------------------------------------------------------------|

|       |        |                                                                                          |
|-------|--------|------------------------------------------------------------------------------------------|
| 2.6.4 | 14 pcs | Eddy coils for bell (1 set = 2 pieces), 7 sets are needed according to WIA specification |
|-------|--------|------------------------------------------------------------------------------------------|

|       |       |                                                                                          |
|-------|-------|------------------------------------------------------------------------------------------|
| 2.6.5 | 4 pcs | Eddy coils for stem (1 set = 2 pieces), 2 sets are needed according to WIA specification |
|-------|-------|------------------------------------------------------------------------------------------|

| Item       | Qty. | Description                                                             |
|------------|------|-------------------------------------------------------------------------|
| <b>2.7</b> |      | <b>Assembly, testing, Run-off, documentation, shipment in EFD China</b> |

|       |   |                                                                                 |
|-------|---|---------------------------------------------------------------------------------|
| 2.7.1 | 1 | Run-off at EFD incl. short term training session (duration 4 working days max.) |
|-------|---|---------------------------------------------------------------------------------|

The end user will provide 50 pieces each kind of part for the purpose of testing the machine

|       |   |                                                                                         |
|-------|---|-----------------------------------------------------------------------------------------|
| 2.7.2 | 1 | Manuals (4 copies each in paper version and electronic version). Labels in English only |
|-------|---|-----------------------------------------------------------------------------------------|

- ✧ operation manual
- ✧ electric manual
- ✧ power source manual
- ✧ maintenance manual
- ✧ lubrication manual

- ✧ cooling circuit manual
- ✧ list of spare parts
- ✧ list of purchased items

2.7.3      1      Packing and truck transportation to Shanghai seaport, FOB Shanghai

| Item | Qty. | Description |
|------|------|-------------|
|------|------|-------------|

---

## 2.8                      **Installation and start up**

|       |   |                                                                                                                                                                                                                                                                                                                             |
|-------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.8.1 | 1 | Supervision of installation at customer's plant including all expenses for travel, lodging, daily allowances etc., calculation based on 5 complete working days for 1 EFD-technician<br><br>Customer should provide one electrician, one mechanical specialist and is responsible for the transport in the customer's plant |
| 2.8.2 | 1 | Run-off at WIA incl. training session (duration 20 working days max.)                                                                                                                                                                                                                                                       |
| 2.8.3 |   | Acceptance Criteria. The machine is pre and final accepted according to the technical agreement signed between the Seller and the buyer.                                                                                                                                                                                    |
| 2.8.4 |   | The seller's engineer provides training for the buy's engineers in terms of operation, mechanical, electric and maintenance.                                                                                                                                                                                                |

### **3. Delivery and Payment Conditions**

#### **3.1. EX WORKS SHIPPING DATE**

7 months after the effectiveness of the commercial contract.

#### **3.2. PRE - ACCEPTANCE TESTS AT SELLER'S WORKS**

Test pieces must be available for machine and process tools approval in sufficient number at supplier's works not later than 8 weeks prior to agreed shipping date. Work pieces must be identical to the parts, which will be used for final acceptance tests at buyer's works.

#### **3.3. PROCESS TOOLS DEMONSTRATION DURING PRE-ACCEPTANCE TESTS AT SELLER'S WORKS AND FINAL ACCEPTANCE TESTS AT BUYER'S WORKS**

Max. 2 types of work pieces will be demonstrated (including make of cuttings, etching works, hardness tests etc.)

If final acceptance at purchaser's works depends on a successful test run under series conditions, purchaser needs to supply work pieces in sufficient number free of charge to the supplier.

If this cannot be arranged, it does not release buyer from giving his final approval resp. from the agreed payment duties.

#### **3.4. PAINT**

RAL 9002

#### **3.5. UTILITIES**

3 x 380V +10%/-10%, 50Hz

#### **3.6. COPY RIGHT**

The specification and technology contained in our proposal are the property of EFD. Passing on to third parties is not permitted as well as utilization of our proposal by buyer himself.

#### **3.7. WARRANTY**

Providing EFD's manual is put into practice, 12 months (three shifts usage) from putting into operation or 15 months after the delivery date, whichever comes first.